Figure 5: **Conditional multi-view generation on a real image.** We can generate arbitrary camera trajectories from a given real image for multi-view synthesis, paving the way to single-image scene reconstruction using camera-controlled video models.

is crucial to incorporate the spatio-temporal Plücker embeddings into the spatio-temporal patches instead of using the context vector.

4.3 Applications

Image-to-video generation. We synthesize camera-controlled videos based on different camera poses as shown in Fig. 1. For this task we use a version of our pre-trained text-to-video model that we fine-tune on video sequences where a random subset of the input frames are masked. At inference time, we can provide image guidance for any of the generated frames, providing an additional dimension of controllability when paired with our proposed method for camera control. To demonstrate image-to-video generation in Fig. 1, we condition the model using camera poses along with image guidance from the first frame of a generated video sequence. Note that our method provides no control over motion within the scene itself; hence, scene motion can differ depending on the random seed or the provided camera poses.

Image-to-multiview generation. We also explore multi-view generation for static scenes as shown in Fig. 5. Given a real input image of a complex scene unseen during training, our camera-conditioned model generates view-consistent renderings of that scene from arbitrary viewpoints. These multi-view renderings could be directly relevant to downstream 3D reconstructions pipelines, e.g., based on NeRF (Mildenhall et al., 2020) or 3D Gaussian Splatting (Kerbl et al., 2023). We show the potential of camera-conditioned image-to-multiview generation for complex 3D scene generation, but we leave more extensive exploration of this topic for future work.

5 Conclusion

Large-scale video transformer models show immense promise to solve many long-standing challenges in computer vision, including novel-view synthesis, single-image 3D reconstruction, and text-conditioned scene synthesis. Our work brings additional controllability to these models, enabling a user to specify the camera poses from which video frames are rendered.

Limitations and future work. There are several limitations to our work, which highlight important future research directions. For example, while rendering static scenes from different camera viewpoints produces results that appear 3D consistent, dynamic scenes rendered from different camera viewpoints can have inconsistent motion (see supplemental videos). We envision that future video generation models will have fine-grained control over both scene motion and camera motion to address this issue. Further, our approach applies camera conditioning only to the low-resolution SnapVideo model and we keep their upsampler model frozen (i.e., without camera conditioning)—it may be possible to further improve camera control through joint training, though this brings additional architectural engineering and computational challenges. Finally, our approach is currently limited to generation of relatively short videos (16 frames), based on the design and training scheme of the SnapVideo model. Future work to address these limitations will enable new capabilities for applications in computer vision, visual effects, augmented and virtual reality, and beyond.

6 Ethics Statement

Broader Impact. Recent video generation models demonstrate coherent, photorealistic synthesis of complex scenes—capabilities that are highly sought after for numerous applications across computer vision, graphics, and beyond. Our key technical contributions relate to camera control of these models, which can be applied to a wide range of methods. As with all generative models and technologies, underlying technologies can be misused by bad actors in unintended ways. While these methods continue to improve, researchers and developers should continue to consider safeguards, such as output filtering, watermarking, access control, and others.

Data. To develop the camera control methods proposed in this paper, we used RealEstate10K (Zhou et al., 2018). RealEstate10K is released and open-sourced by Google LLC under a Creative Commons Attribution 4.0 International License, and sourced from content using a CC-BY license. The dataset can be found under the following URL: `https://google.github.io/realestate10k`. The RealEstate10K dataset was published as part of a research paper by Zhou et al. (2018): "Stereo Magnification: Learning View Synthesis using Multiplane Images" at SIGGRAPH 2018. The RealEstate10K dataset is a commonly used dataset for 3D reconstruction (Charatan et al., 2024), 3D generation (Gao et al., 2024c), and camera-controlled video generation (Wang et al., 2023e). Similarly, closely related baselines such as MotionCtrl (Wang et al., 2023e) and CameraCtrl (He et al., 2024a) use the identical dataset to train and evaluate their camera-controlled video diffusion models. Hence, we follow an established training and evaluation pipeline without any modifications to the data.

7 Reproducibility Statement

We have structured our paper to ensure comprehensive reproducibility of our camera control method. Section 3 provides a detailed description of our method, including theoretical foundations and core algorithmic components. Section 4 presents thorough experimental details, covering evaluation protocols, metrics, and comparisons with baselines. All implementation specifics, including training procedures, architectural choices, and hyperparameters, are documented in Appendix C. To further facilitate reproducibility, we include the source code of our camera-controlled FIT block as supplementary material. The provided implementation contains the core components necessary to replicate our approach, accompanied by documentation and usage examples. We welcome requests for additional technical details or clarifications to support the research community in reproducing and building upon our work.

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